

Why Can't We Make Qubits with Only Capacitors and Linear Inductors?

1 Introduction

A quantum bit (qubit) is a quantum system that has two well-separated energy levels that can be individually addressed while remaining isolated from all other energy levels.

A natural question is:

Why can't we simply build a qubit using only capacitors and linear inductors?

Although capacitors and inductors are the fundamental building blocks of superconducting circuits, they are *not sufficient* by themselves to create a useful qubit. The reason lies in the energy spectrum of an LC circuit.

This document explains the reason step by step.

2 Step 1: Consider a Simple LC Circuit

An ideal LC circuit consists of

- a capacitor C
- an inductor L

The classical energy stored in the circuit is

$$H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L},$$

where

- Q is the charge on the capacitor,
- Φ is the magnetic flux through the inductor.

Notice that both terms are quadratic.

3 Step 2: Quantize the Circuit

To obtain the quantum description, we promote the variables to operators satisfying

$$[\hat{\Phi}, \hat{Q}] = i\hbar.$$

The Hamiltonian becomes

$$\hat{H} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}.$$

This is mathematically identical to the Hamiltonian of the quantum harmonic oscillator.

4 Step 3: Solve for the Energy Levels

The energy eigenvalues of the quantum harmonic oscillator are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

where

$$\omega = \frac{1}{\sqrt{LC}}.$$

The first few levels are

$$E_0 = \frac{1}{2}\hbar\omega,$$

$$E_1 = \frac{3}{2}\hbar\omega,$$

$$E_2 = \frac{5}{2}\hbar\omega,$$

and so on.

The spacing between adjacent levels is

$$E_{n+1} - E_n = \hbar\omega.$$

This spacing is constant.

5 Step 4: Why Equal Spacing is a Problem

Suppose we want to define

$$|0\rangle = \text{ground state},$$

$$|1\rangle = \text{first excited state}.$$

To manipulate the qubit, we apply microwave radiation at frequency

$$f = \frac{\omega}{2\pi}.$$

However,

$$E_1 - E_0 = E_2 - E_1 = E_3 - E_2 = \hbar\omega.$$

Therefore the microwave field that drives

$$|0\rangle \rightarrow |1\rangle$$

also drives

$$|1\rangle \rightarrow |2\rangle,$$

and

$$|2\rangle \rightarrow |3\rangle.$$

The system behaves like a ladder with equally spaced rungs.
As a result,

- population leaks into higher excited states,
- it is impossible to isolate only two levels,
- the circuit does not behave as a qubit.

6 Step 5: The Need for Anharmonicity

A useful qubit requires that

$$E_1 - E_0 \neq E_2 - E_1.$$

This property is called **anharmonicity**.

Anharmonicity ensures that a microwave pulse resonant with

$$|0\rangle \rightarrow |1\rangle$$

is off resonance for

$$|1\rangle \rightarrow |2\rangle.$$

Consequently,

- only two levels participate,
- leakage is minimized,
- coherent quantum control becomes possible.

7 Step 6: Why Capacitors and Linear Inductors Cannot Produce Anharmonicity

A capacitor stores energy

$$U_C = \frac{Q^2}{2C},$$

which is quadratic.

A linear inductor stores energy

$$U_L = \frac{\Phi^2}{2L},$$

which is also quadratic.

Adding quadratic functions together still produces a quadratic potential,

$$U = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}.$$

Quadratic potentials always correspond to harmonic oscillators.

Therefore,

Linear capacitors + linear inductors $\implies$ harmonic oscillator

and harmonic oscillators always have equally spaced energy levels.

Thus,

No anharmonicity $\implies$ No qubit

8 Step 7: What is Added in Real Superconducting Qubits?

To break the equal spacing, superconducting qubits include a nonlinear element.

The most common nonlinear element is the Josephson junction.

Its potential energy is

$$U(\phi) = -E_J \cos \phi,$$

where

- E_J is the Josephson energy,
- ϕ is the superconducting phase difference.

Unlike a quadratic potential, the cosine potential is nonlinear.

This produces unequally spaced energy levels,

$$E_1 - E_0 \neq E_2 - E_1.$$

Now a microwave pulse can selectively excite only

$$|0\rangle \leftrightarrow |1\rangle,$$

without significantly exciting higher states.

This is precisely why Josephson junctions are the key ingredient in superconducting qubits such as the transmon, flux qubit, and phase qubit.

9 Summary

The reason an LC circuit cannot function as a qubit can be summarized as follows:

1. Capacitors and linear inductors produce a quadratic Hamiltonian.
2. Quantization yields a quantum harmonic oscillator.
3. Harmonic oscillators have equally spaced energy levels.
4. Microwave control cannot isolate only the lowest two states because all adjacent transitions occur at the same frequency.
5. A qubit requires anharmonic energy levels.
6. Anharmonicity is introduced by a nonlinear element, typically a Josephson junction.

Hence,

$\text{Capacitors} + \text{Linear Inductors} \not\Rightarrow \text{Qubit}$

while

$\text{Capacitors} + \text{Linear Inductors} + \text{Josephson Junction} \Rightarrow \text{Superconducting Qubit.}$